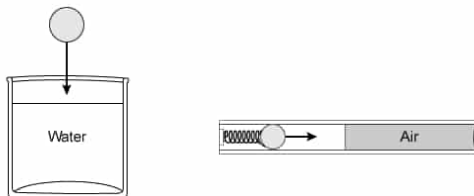


Students investigating the motion of golf balls in a medium perform two experiments where golf balls are introduced into different media. In the first experiment, the balls are dropped vertically downward into a tank of water. In the second experiment, the balls are compressed against an ideal spring and then propelled horizontally into a tube filled with air, as shown in the figure below. As the golf ball moves through a medium, it experiences a drag force F_D that opposes its motion, as described by the equation:

$$F_D = \frac{1}{2} C_D \rho A v^2$$

where C_D is the drag coefficient, ρ is the medium's density, A is the cross-sectional area of the ball, and v is the velocity of the ball.



The students compress a 40 g golf ball a distance of 80 cm against a spring with spring constant $k = 100$ N/m. After it is released from the spring, what is the velocity of the ball as it enters the air-filled tube?

- ✓ ☒ A. 40 m/s [53%]
☐ B. 45 m/s [19%]
☐ C. 56 m/s [20%]
☐ D. 125 m/s [6%]

Correct

52% answered correctly

Explanation:

Applying conservation of mechanical energy E to determine the speed of a ball released from a spring

$PE_{el} = \frac{1}{2} kx^2$	• PE_{el} Elastic potential energy	• KE Kinetic energy of golf ball
$KE = \frac{1}{2} mv^2$	• k Spring constant	• m Mass of golf ball
	• x Displacement from equilibrium	• v Speed of golf ball

Ball compressed 80 cm against spring

$E_{initial} = PE_{el} = \frac{1}{2} kx^2$

Ball released, gains KE

$E_{final} = KE = \frac{1}{2} mv^2$

Conservation of energy: $E_{initial} = E_{final} \Rightarrow v = \sqrt{\frac{k}{m} x} = \sqrt{\frac{100 \text{ N/m}}{0.04 \text{ kg}} (0.8 \text{ m})} = 40 \frac{\text{m}}{\text{s}}$

When a system consists of an object attached to a spring, the compression or stretching of the spring causes **elastic potential energy** PE_{el} to be stored in the system. This PE_{el} is equal to half the product of the spring constant k and the square of the distance x that the spring is compressed or stretched from equilibrium (ie, its uncompressed or unstretched length):

$$PE_{el} = \frac{1}{2} kx^2$$

Consequently, the total **mechanical energy** E of the system is the sum of the object's **kinetic energy** KE and PE_{el} :

$$E = KE + PE_{\text{el}}$$

$$E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

The E of the object-spring system is conserved (ie, constant) when there are no external forces (eg, friction) acting on the system. Hence, **conservation of energy** implies that the E_{initial} of the system must be equal to its E_{final} :

$$E_{\text{initial}} = E_{\text{final}}$$

In this question, when the golf ball is compressed against the spring, the E_{initial} of the system equals PE_{el} because the ball has zero initial speed:

$$E_{\text{initial}} = \frac{1}{2}kx^2$$

When the spring is released, the PE_{el} is transformed entirely into kinetic energy and E_{final} equals KE :

$$E_{\text{final}} = \frac{1}{2}mv^2$$

Conservation of energy implies that:

$$\frac{1}{2}kx^2 = \frac{1}{2}mv^2$$

Solving this equation for the speed v yields the following equation in terms of k , x , and the ball's mass m :

$$v = \sqrt{\frac{k}{m}}x$$

Inserting $k = 100 \text{ N/m}$, $m = 40 \text{ g}$, and $x = 80 \text{ cm}$ into the equation gives:

$$v = \sqrt{\frac{100 \frac{\text{N}}{\text{m}}}{40 \text{ g}}}(80 \text{ cm})$$

Converting to **SI units** and simplifying using the equivalent expression of the **derived unit** yields:

$$v = \sqrt{100 \frac{\text{kg} \cdot \text{m}}{0.04 \text{ kg}}}(0.8 \text{ m})$$

$$v = \sqrt{2,500} \cdot (0.8) \frac{\text{m}}{\text{s}}$$

$$v = 40 \frac{\text{m}}{\text{s}}$$

(Choice B) 45 m/s results from neglecting to square the distance x in the formula for spring potential energy.

(Choice C) 56 m/s results from neglecting the overall factor of $\frac{1}{2}$ in the formula for spring potential energy.

(Choice D) 125 m/s follows from neglecting to convert the mass m of the ball and distance x into SI units.

Educational objective:

When a compressed spring is released with an object pressing against it, the elastic potential energy of the spring is transferred to the kinetic energy of the object. Conservation of energy can be used to determine the final speed of the object.

Time spent:0 sec

Subject:

Physics

QID:403219

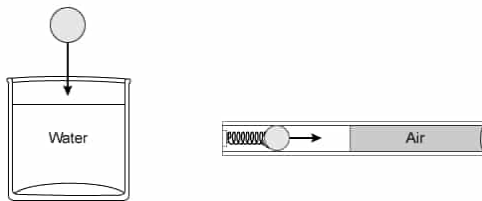
System:

Mechanics and Energy

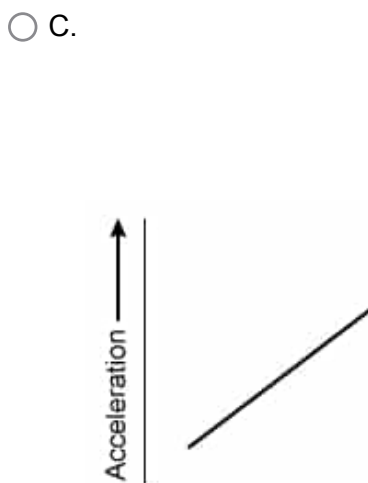
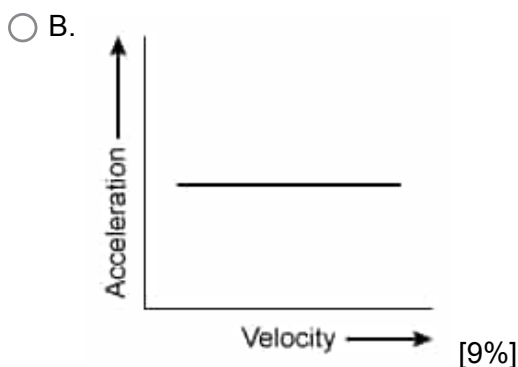
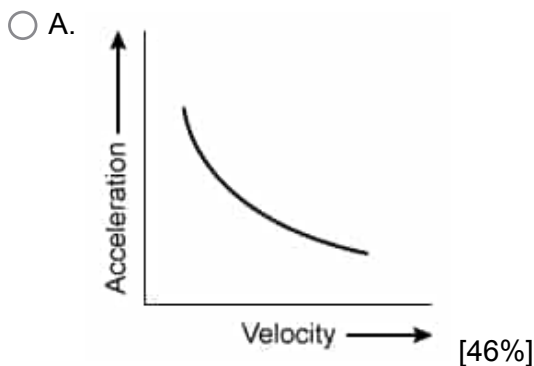
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$$F_D = \frac{1}{2}C_D\rho Av^2$$

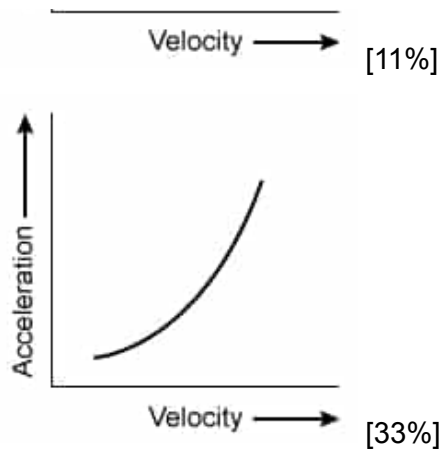
where C_D is the drag coefficient, ρ is the medium's density, A is the cross-sectional area of the ball, and v is the velocity of the ball.



Which of the following graphs best illustrates how the magnitude of the golf ball's acceleration varies with its velocity in the air-filled region of the tube?



✓ ☒ D.

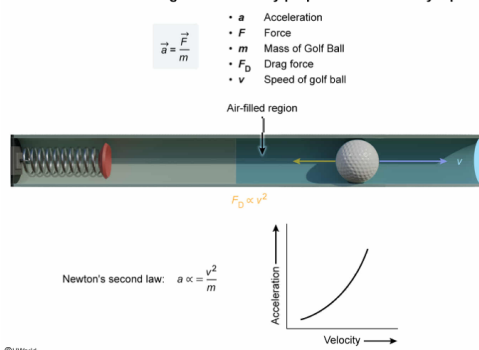


Correct

32% answered correctly

Explanation:

Acceleration due to a drag force is directly proportional to velocity squared



Newton's second law of motion states that the force F exerted on an object is equal to the product of the object's mass m and its **acceleration** a :

$$\vec{F} = m \vec{a}$$

Solving this equation for acceleration yields:

$$\vec{a} = \frac{\vec{F}}{m}$$

In this question, the golf ball experiences a drag force F_D that opposes its motion as it travels through the air-filled region. Consequently, Newton's second law implies that the magnitude of the ball's acceleration is the ratio of F_D and the ball's mass m :

$$a = \frac{F_D}{m}$$

Inserting the formula for F_D given in the passage yields:

$$a = \frac{\frac{1}{2} C_D \rho A v^2}{m}$$

Hence, the magnitude of acceleration is **directly proportional** to the *square* of the ball's velocity v :

$$a \propto v^2$$

Therefore, the graph in Choice D that shows a nonlinear increase in acceleration as velocity increases best represents the behavior of the golf ball's acceleration as a function of velocity.

(Choice A) This graph represents an acceleration that decreases as velocity increases, which is inconsistent with the formula for drag force.

(Choice B) This graph represents an acceleration that is independent of velocity, but drag force increases as velocity increases.

(Choice C) This graph represents an acceleration that increases linearly as velocity increases. However, the acceleration due to drag increases nonlinearly with the *square* of velocity.

Educational objective:

An object's acceleration is equal to the net force exerted on the object divided by the object's mass.

Time spent:0 sec

Subject:

Physics

QID:403220

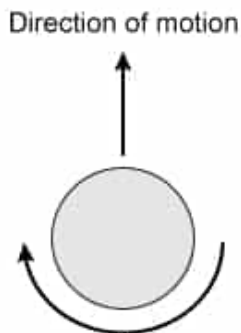
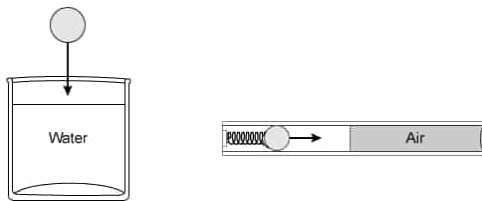
System:

Mechanics and Energy

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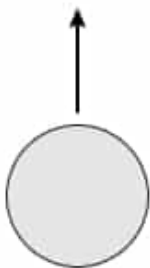
$$F_D = \frac{1}{2} C_D \rho A v^2$$

where C_D is the drag coefficient, ρ is the medium's density, A is the cross-sectional area of the ball, and v is the velocity of the ball.



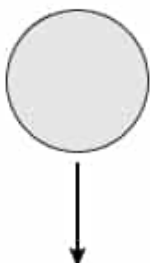
The motion of spinning objects is also affected by the fluid medium. A golf ball with clockwise sidespin about its axis (when viewed facing the direction of travel) exerts a leftward force on the surrounding air. Which image best illustrates the corresponding force the air exerts back on the golf ball?

☐ A.



[1%]

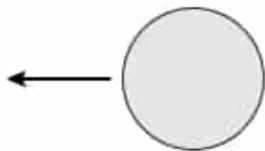
☐ B.



[100%]

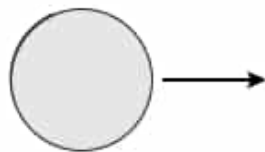
[18%]

○ C.



[9%]

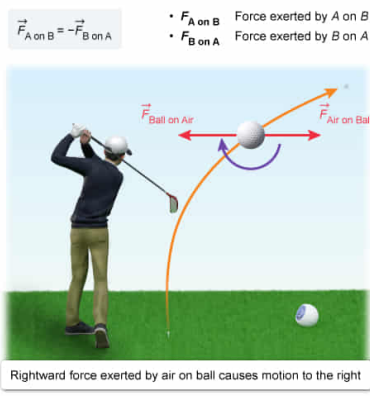
✔ ○ D.



[70%]

Correct

70% answered correctly

Explanation:**Applying Newton's third law to the motion of a spinning golf ball**

According to [Newton's third law of motion](#) a pair of **forces** exists when two objects (*A* and *B*) interact. When object *A* exerts a force F on object *B* (ie, the action force), object *B* exerts an equal and opposite force back on object *A* (ie, the reaction force):

$$\vec{F}_{A \text{ on } B} = - \vec{F}_{B \text{ on } A}$$

In this question, the spinning golf ball exerts a net leftward force on the surrounding air medium. Newton's third law implies that the air must exert an equal and opposite net *rightward force* on the golf ball:

$$\vec{F}_{\text{Air on Ball}} = - \vec{F}_{\text{Ball on Air}}$$

Therefore, the image that best illustrates the force on the golf ball is shown in Choice D.

(Choices A and B) These images show upward and downward forces on the golf ball, which are inconsistent with Newton's third law.

(Choice C) Because the ball exerts a leftward force on the surrounding air, the force exerted by the air on the ball must be opposite (ie, rightward).

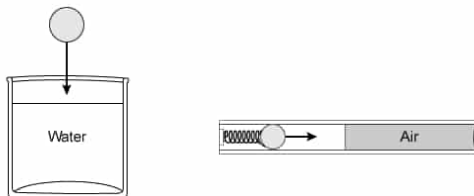
Educational objective:

Newton's third law of motion implies that forces exist in pairs. When an object exerts a force on a second object, the second object exerts an equal and opposite force back on the first object.

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$$F_D = \frac{1}{2} C_D \rho A v^2$$

where C_D is the drag coefficient, ρ is the medium's density, A is the cross-sectional area of the ball, and v is the velocity of the ball.



The water in the tank is replaced by a fluid with a density greater than the average density of a golf ball. Which of the following will occur when the golf ball is placed in the new fluid?

- ✓ ☒ A. The golf ball will float on the surface of the fluid. [88%]
☐ B. The golf ball will be suspended in the fluid. [4%]
☐ C. The golf ball will briefly float on the surface and then sink to the bottom of the tank. [2%]
☐ D. The golf ball will immediately sink to the bottom of the tank. [4%]

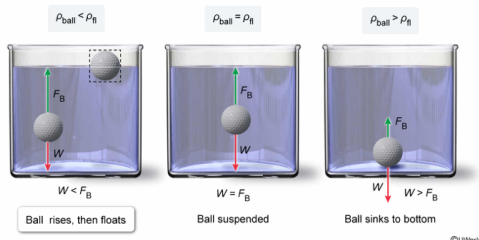
Correct

87% answered correctly

Explanation:

A golf ball floats when its density is less than the density of the fluid

- ρ_{ball} Density of golf ball
- ρ_{fl} Density of fluid
- W Weight of golf ball
- F_B Buoyant force



Archimedes' principle states that an object in a fluid experiences a net upwardly directed **buoyant force** F_B equal to the product of the fluid's density ρ_{fl} , the gravitational acceleration g , and the volume V_{fl} of the fluid displaced by the object:

$$F_B = \rho g V_{\text{fl}}$$

Furthermore, the object's **weight** W equals the product of the object's **density** ρ_{obj} , g , and the volume of the object V_{obj} :

$$W = \rho_{\text{obj}} g V_{\text{obj}}$$

When an object is fully immersed in the fluid, $V_{\text{fl}} = V_{\text{obj}}$ and the object's behavior is determined by the relationship between ρ_{obj} and ρ_{fl} . When $\rho_{\text{obj}} > \rho_{\text{fl}}$, the weight W of the object is greater than F_B and the object will immediately sink in the fluid:

$$\rho_{\text{obj}} > \rho_{\text{fl}} \quad \Rightarrow \quad W > F_B$$

When $\rho_{\text{obj}} = \rho_{\text{fl}}$, then W equals F_B and the object will be suspended in the fluid:

$$\rho_{\text{obj}} = \rho_{\text{fl}} \quad \Rightarrow \quad W = F_B$$

When $\rho_{\text{obj}} < \rho_{\text{fl}}$, then W is less than F_B and the object will rise in the fluid until it floats on the surface:

$$\rho_{\text{obj}} < \rho_{\text{fl}} \quad \Rightarrow \quad W < F_B$$

In this question, the average density of the golf ball is less than the fluid in the tank. Therefore, the ball will float on the surface of the fluid.

(Choice B) The golf ball will remain suspended in the fluid only if the density of the fluid *equals* the density of the ball.

(Choice C) When the golf ball's density is less than the density of the fluid, the ball will float. The ball will sink only if sufficient weight is added to the ball to overcome F_B .

(Choice D) The golf ball will sink to the bottom only if the density of the ball is *greater than* the density of the fluid.

Educational objective:

The behavior of an object in a fluid is determined by the relationship between the densities of the object and the fluid. An object floats when its density is less than the density of the fluid.

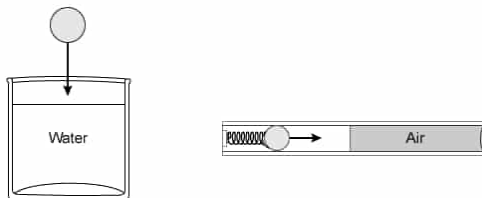
Time spent: 0 sec
Subject:
Physics

QID: 403222
System:
Mechanics and Energy

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$$F_D = \frac{1}{2}C_D\rho Av^2$$

where C_D is the drag coefficient, ρ is the medium's density, A is the cross-sectional area of the ball, and v is the velocity of the ball.



In the first experiment, which of the following best describes the vertical motion of the golf ball through the water when the drag force, the buoyant force, and weight of the ball sum to zero?

- ☐ A. It accelerates downward. [9%]
- ☐ B. It accelerates upward. [6%]
- ☒ C. It moves with constant speed downward. [73%]
- ☐ D. It moves with constant speed upward. [10%]

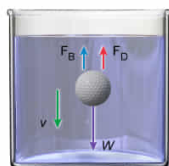
Correct

73% answered correctly

Explanation:

Applying Newton's first law to motion in a fluid medium

- F_{net} Net force
- F_B Buoyant force
- F_D Drag force
- W Weight



As ball falls in water, F_D increases until F_{net} is zero

$$F_{\text{net}} = F_B + F_D - W = 0$$

Zero F_{net} implies that the ball moves downward at constant velocity v in the water

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Newton's first law of motion implies that objects with **inertia** (ie, mass) resist changes in their velocity. Hence, objects at rest remain at rest and objects in motion remain in motion with constant velocity unless a **net external force** is exerted upon them.

In this question, the golf ball entering the tank of water experiences three forces: the upward **buoyant force** F_B due to the displaced water, the upward drag force F_D opposing the ball's downward motion, and the ball's downward **weight** W . As the ball falls in the water, F_D increases until the sum of all the forces

on the ball (ie, the net force) is zero:

$$F_{\text{net}} = F_B + F_D - W = 0$$

When the net force is zero, Newton's first law implies that the golf ball must move at a constant velocity (ie, zero acceleration). Therefore, because the ball was initially dropped downward, it continues to move downward at a constant velocity in the water.

(Choices A and B) The golf ball only **accelerates** when a **nonzero net force** is exerted upon it.

(Choice D) The golf ball cannot move upward at constant velocity because it was initially dropped downward.

Educational objective:

When zero net force is exerted on an object, the object must be at rest or moving at a constant velocity.

Time spent:0 sec

Subject:
Physics

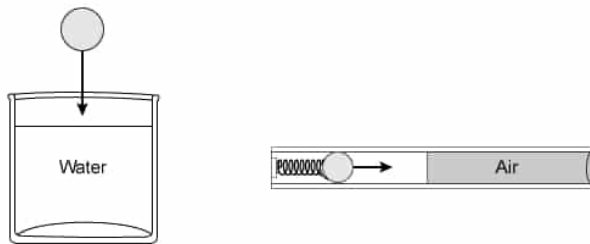
QID:403223

System:
Mechanics and Energy

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$$F_D = \frac{1}{2}C_D\rho Av^2$$

where C_D is the drag coefficient, ρ is the medium's density, A is the cross-sectional area of the ball, and v is the velocity of the ball.



Time (s)	Velocity (m/s)
1	15
2	25
3	35
4	40
5	40

The table above shows the velocity of a golf ball measured at different times after it is dropped downward from a great height. How did the ball's acceleration vary with time?

- ☐ A. It was constant in time [4%]
- ✓ ☒ B. It was first constant, then decreased with time [85%]
- ☐ C. It was first constant, then increased with time [1%]
- ☐ D. It increased with time [7%]

Correct

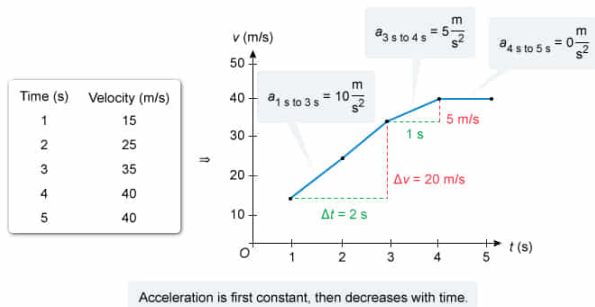
85% answered correctly

Explanation:

Determining acceleration from a table of velocities at given times

$$\vec{a} = \frac{\Delta \vec{v}}{\Delta t}$$

- $\vec{a}$ Acceleration
- $\Delta \vec{v}$ Change in velocity
- Δt Time interval



An object's **acceleration** a is the change in its **velocity** Δv over an interval of time Δt :

$$\vec{a} = \frac{\Delta \vec{v}}{\Delta t}$$

In this question, the golf ball's velocity v is measured at 1 s intervals as it falls. The acceleration during each of these intervals can be determined by subtracting the initial velocity v_i from the final velocity v_f :

$$\Delta v = v_f - v_i$$

Hence, from 1 s to 2 s, the ball's acceleration equals:

$$a_{1 \text{ s to } 2 \text{ s}} = \frac{25 \frac{\text{m}}{\text{s}} - 15 \frac{\text{m}}{\text{s}}}{1 \text{ s}} = 10 \frac{\text{m}}{\text{s}^2}$$

In the intervals from 2 s to 3 s and then from 3 s to 4 s, a equals:

$$a_{2 \text{ s to } 3 \text{ s}} = \frac{35 \frac{\text{m}}{\text{s}} - 25 \frac{\text{m}}{\text{s}}}{1 \text{ s}} = 10 \frac{\text{m}}{\text{s}^2}$$

$$a_{3 \text{ s to } 4 \text{ s}} = \frac{40 \frac{\text{m}}{\text{s}} - 35 \frac{\text{m}}{\text{s}}}{1 \text{ s}} = 5 \frac{\text{m}}{\text{s}^2}$$

Finally, from 4 s to 5 s, a is zero:

$$a_{4 \text{ s to } 5 \text{ s}} = \frac{40 \frac{\text{m}}{\text{s}} - 40 \frac{\text{m}}{\text{s}}}{1 \text{ s}} = 0 \frac{\text{m}}{\text{s}^2}$$

Therefore, the acceleration is constant at first (ie, from 1 to 3 s), then it decreases with time (from 3 s to 5 s).

(Choice A) The acceleration is not constant with time because the differences in velocities between each 1 s time interval are not all the same.

(Choice C) The acceleration is initially constant but after 3 s, it begins to *decrease* as the velocity approaches 40 m/s.

(Choice D) The velocity initially increases, but the acceleration is initially constant and then begins to decrease.

Educational objective:

An object's acceleration can be calculated by determining the change in its velocity over a time interval.

Time spent:0 sec

Subject:
Physics

QID:403224

System:
Mechanics and Energy